

A CLASSICAL READING OF THE COSMETIC-SURGERY ARGUMENT

Status. This note reorganizes the manuscript's argument in the language of Floer, Donaldson, Kronheimer–Mrowka, Feehan–Leness and DLME. It has not been checked against the manuscript line by line. Labels: [V] means checked by hand or by machine; [P] means plausible; [S] means speculative. Supporting notes are in `notes/`. Throughout, $Y_r = S_r^3(K)$, and DLME means Daemi–Lidman–Miller Eismeier, arXiv:2410.21248.

1. The theorem and the shape of the argument

By Hanselman and DLME (Cor. 1.4), a purely cosmetic pair must be $\{\pm 2\}$ with $g(K) = 2$. The method proves the following more symmetric statement (`notes/T4`).

Theorem 1 (conditional on Theorem 4). *Let $K \subset S^3$ be nontrivial. There do not exist negative-definite cobordisms $V: Y_{-2} \rightarrow Y_2$ and $V': Y_2 \rightarrow Y_{-2}$ with $b_1 = 0$, both inducing isomorphisms on $I(\cdot; \mathbb{F}_2)$, such that $\pi_1(V')$ is normally generated by the image of one end. In particular the purely cosmetic surgery conjecture holds.*

The argument has three steps.

- (1) *Algebra.* Surgery exact triangles yield integral maps $B: I(Y_2) \rightarrow I(Y_{-2})$ and $H: I(Y_{-2}) \rightarrow I(Y_2)$ that are isomorphisms mod 2. Together with V they give an automorphism w of the 2-adic lattice of $I(Y_2)$. Relative invariants of two caps give a nonzero pairing.
- (2) *Nonvanishing.* A suitable power w^M is the identity mod 2^N . So a *secondary* (family-of-metrics) relative Donaldson invariant $\Omega(X_M)$ of the capped composite is $\equiv q \not\equiv 0 \pmod{2^N}$.
- (3) *Vanishing.* Once the coupled Dirac index satisfies $n_D \geq 1$, an $\mathrm{SO}(3)$ -monopole (Pidstrigach–Tyurin, Feehan–Leness) relation expresses $2^{n_D-1}\Omega$ through Seiberg–Witten-type reducibles. A family adjunction inequality shows there are none. Hence $\Omega = 0$, a contradiction.

The cosmetic map is used to make the (negative-definite) cobordism V compose with itself. This gives many (-4) -spheres per unit of b^+ .

2. Surgery triangles

We use Floer's irreducible instanton homology with determinant-one gauge group. $Y_{\pm 1}$, $Y_{\pm 2}$ and S^3 carry only central reducibles, $I(S^3) = 0$, and Y_0 carries w with $w \cdot A$ odd. The following dictionary is in `notes/T5`; it is [V] for the topology and [P] for the analysis.

- $f_+: I^w(Y_0) \rightarrow I(Y_2)$ and $f_-: I(Y_{-2}) \rightarrow I^w(Y_0)$ are composites of Floer-triangle maps. They are isomorphisms because the S^3 term vanishes.
- $g_{\pm}$ is DLME's family map g_1 for the triads $\{2, \infty, 0\}$ and $\{0, \infty, -2\}$, with middle term $I(S^3) \oplus I(S^3) = 0$. Their h -maps give $f_+g_+ \simeq 1$ and $g_-f_- \simeq 1$.
- *The distance-four map.* Let $W' = (-X_2(K)) \cup_{S^3} X_{-2}(K): Y_2 \rightarrow Y_{-2}$. Then $b^+ = 0$ and $H_2 = \langle F_l, F_r \rangle \cong \langle -2 \rangle^2$. The union of the two cores is a sphere $S = F_l - F_r$ with $S^2 = -4$, and $\partial \nu S = L(4, 1)$. Let Φ_0 be the map defined by the one-parameter family of metrics running from the S^3 -splitting to the $L(4, 1)$ -splitting, with Donaldson's class $\mu(S)$ inserted.

Proposition 2 (`notes/T2`, `notes/T5`).

- (1) *The class $\mathrm{PD}(S) = 2y$ is even. So the bundles c and $c + \mathrm{PD}(S)$ are the same $\mathrm{SO}(3)$ -bundle, with both end identifications twisted by the flat $\mathbb{Z}/2$ character. Writing ι for this twist, the map*

$$B = \Phi_0 - s \iota \Phi_0 \iota$$

is an integral chain map. The $L(4, 1)$ -wall (minimal trace-zero cap, $\kappa = \frac{1}{4}$) lies in the opposite eigenspace of ι -conjugation and cancels [P].

- (2) *The insertion of $\mu(S)$ is forced. The minimal $L(4, 1)$ -cap has framed index 2, against 0 for the $\mathbb{RP}^3$ -cap of DLME; equivalently, it is forced by degree matching with g_-g_+ [V].*
- (3) *$B \simeq g_-g_+ = (f_+f_-)^{-1} =: H^{-1}$ on homology mod 2, up to a degree-four twist [P], about 80%.*

Remark 3. The manuscript's insertion (normalized holonomy over a sweep of loops across S equal to -1) is $\pm \mu(S)$. The class is c_1 of the determinant line of $\bar{\partial}$ on $S \cong \mathbb{P}^1$, and its canonical section vanishes on the jumping-line divisor $V_S = \{E|_S \not\cong \mathcal{O} \oplus \mathcal{O}\}$ [V]. The manuscript's zero-winding modification is an artifact of holonomy representatives: V_S avoids every reducible with $v \cdot S = 0$ automatically.

Relation to DLME's remark on ± 2 . In DLME's bookkeeping, B has Chern–Simons level minus degree/8 equal to $+\frac{1}{8} - \eta$, which is unfavourable; DLME's g_1 at ± 1 has $-\frac{1}{8} - \eta$. The difference is exactly the forced $\mu(S)$. So B is DLME's “trivial middle term, chain map, unfavourable filtration” variation, and the Chern–Simons filtration alone cannot conclude [V].

3. Relative invariants

Take a symplectic closure $X_0 \supset Y_0$ (Gabai, Eliashberg–Thurston, Eliashberg caps, Kronheimer–Mrowka) with nonzero Donaldson invariant $D_{X_0}^w(x^m h^a)$, $h \cdot A = 1$. Insert copies of the bridge $T = f_- \circ V' \circ f_+$, which is a rational isomorphism of $I^w(Y_0)$. Carry the surface classes across T by a nilpotent deformation over $\mathbb{Q}[t_i]/(t_i^2)$. By Cayley–Hamilton, some T^k with $k \geq 1$ preserves nonvanishing. Cutting along an interior Y_2 gives relative invariants

$$\Psi_- \in I(Y_2; \mathbb{Z}), \quad \Psi_+ \in I(-Y_2; \mathbb{Z}), \quad \langle \Psi_+, \Psi_- \rangle = q \neq 0,$$

from manifolds $C_\pm$ with $b^+ > 0$, each blown up once more [P]. Genus two of K is not used here; the tight genus-two surfaces come from the Lefschetz pencil [P].

4. The secondary invariant and its nonvanishing

Let $w = (VB)^3 \circ V(HB)$, a self-cobordism of Y_2 built from four copies of W' , three copies of V and one copy of W_H . Let X_M be the closed manifold obtained by capping $C_- \cup (\text{pads}) \cup w^M \cup (\text{pads}) \cup C_+$.

- *Topology* [V]. X_M contains $n \approx 4M$ disjoint (-4) -spheres S_i and $m = M$ positive pieces. Each positive piece is $P \cong \mathbb{CP}^2 \# 5\overline{\mathbb{CP}}^2$ for every K , contains a square-zero sphere U , and contains the plumbing $(-S_j, U, S_{j+1})$ with squares $(-4, 0, -4)$. Since all the W' pass through S^3 , X_M is a connected sum along those spheres. Its ordinary Donaldson and Seiberg–Witten invariants therefore vanish.
- *The secondary invariant* (notes/T1, [P]). $\Omega(X_M)$ is the count of instantons over the cube $[0, 1]^n$ of metrics whose faces stretch the necks S^3 and $L(4, 1)$, cut down by $\mu(S_1) \cdots \mu(S_n)$ and the cap classes, and summed with signs over the bundles $w_0 + \sum e_i \text{PD}(S_i)$. Single-bundle counts are not invariant; the signed sum is. By gluing, $\Omega(X_M) = \pm \langle \Psi_+, w^M \Psi_- \rangle$.
- *Nonvanishing* [V] as algebra. Each factor of w is an isomorphism mod 2, so w is invertible on $L \otimes \mathbb{Z}_{(2)}$, where $L = I(Y_2; \mathbb{Z})/\text{Tor}$. Choose N with $2^N \nmid q$ and take M divisible by the exponent of $\text{GL}_r(\mathbb{Z}/2^N)$. Then $\Omega(X_M) \equiv \pm q \not\equiv 0 \pmod{2^N}$.

5. Coupling to a spin^c structure

Fix a characteristic class l and put $\Lambda = l + c$, $\Theta = (\Lambda^2 - \sigma)/4$ and $\kappa = c_2 - c^2/4$. The coupled Dirac operator on $W^+ \otimes E$ has complex index $n_D = \Theta - \kappa$ on closed manifolds. For an instanton A from α to α' on a cobordism, APS gives [V] (notes/B1)

$$n_D(A) = \Theta_W - E(A) + \rho_t(\alpha) - \rho_{t'}(\alpha'), \quad \rho_t(\alpha) = \frac{1}{2}(\eta(D_{\alpha,t}) - h) + \frac{1}{4}\eta_{\text{sign}}(Y).$$

Hence $\text{CS} - \rho_t$ is integer-valued up to a constant, and $8(\text{CS} - \rho_t) - \text{gr}$ is an integer grading from which CS cancels. The “spin-filtered” Floer group is really a *spin-graded* one [V].

On X_M the telescope gives $\Theta = m + \Theta_0$ and $8\kappa = n + 3m + c$, so

$$n_D(X_M) = \frac{5m - n}{8} + c_1.$$

Each copy of W' lowers n_D by $\frac{1}{8}$ (one parameter, one degree-two insertion, $\Theta = 0$). Each positive piece raises it by $\frac{5}{8}$ ($b^+ = 1$ costs $\frac{3}{8}$, and $\Theta = 1$ gains 1) [V].

6. The secondary Feehan–Leness relation

Theorem 4 (notes/T1, Thm. A; [P], about 70%). *Suppose $n_D \geq 1$. After $n_D - 1$ cuts by sections of the phase-weight-two line, the free part of the cut-down $\text{SO}(3)$ -monopole moduli space over the cube is a compact oriented 1-manifold, possibly with rational multisection weights. Its ends are:*

- (1) *the instanton links $\mathbb{CP}^{n_D-1}$, each cut to 2^{n_D-1} points, with total $2^{n_D-1}\Omega$;*
- (2) *lens-face ends, which cancel in pairs of bundles c , $c + \text{PD}(S_i)$;*
- (3) *ends at abelian limits: Seiberg–Witten classes $K = \Lambda + v$ at Feehan–Leness level $\ell(v) = \kappa + v^2/4 \geq 0$, broken limits, and mixed limits containing zero-spinor abelian segments.*

Ends at the S^3 -faces without an abelian piece vanish for dimensional reasons. Hence $2^{n_D-1}\Omega = -\#\{\text{abelian ends}\}$.

Inputs.

- Standard: Feehan–Leness compactness, transversality and instanton links; the Pidstrigach–Tyurin structure; Morgan–Mrowka–Ruberman and Donaldson for psc necks with stabilizers; weighted Stokes.
- New: the lens cancellation carried into the coupled problem (the Dirac determinant is complex and adds no sign; about 70%). The treatment of mixed limits by projecting away abelian zero-spinor segments (the manuscript’s “necessary projection”; 55–60%) is the single genuinely new analytic idea. It replaces Feehan–Leness’s obstructed gluing at reducibles.

7. The Seiberg–Witten side is empty

Three classical facts enter.

- (1) *μ -classes on reducibles* (notes/T2, [V]). At $E = L_1 \oplus L_2$ with $v = c_1(L_1) - c_1(L_2)$, $\mu(S)$ restricts to the link as $\pm \frac{v \cdot S}{2} h$, and V_S contains the reducible if and only if $v \cdot S \neq 0$. So an insertion is met only if $v \cdot S \neq 0$, which is even. Otherwise it requires a bubble on S , i.e. a unit of Feehan–Leness level, with distinct bubbles for disjoint spheres.
- (2) *$b^+ = 1$ chambers on rational-surface pieces* (notes/T3). Type-S solutions on P solve the Seiberg–Witten equations for K perturbed by $\beta^+ \in 2\pi\Lambda$.
 - A family of toric Kähler metrics with $\text{Scal} \geq 0$ (Guillemin, Abreu) with period $H = U + aX_1 + bX_2$, $a, b \in [0, \frac{1}{4}]$, gives by Weitzenböck the band $(v \cdot H)(K \cdot H) < 0$. Since $2\chi + 3\sigma = 4 > 0$ there are no walls at zero perturbation, and Kronheimer–Mrowka/Li–Liu wall-crossing shows the band is optimal [V].
 - Convexity over the vertex classes $H_I = U + \frac{1}{4} \sum_{i \in I} X_i$ yields the chamber condition [V]. Each H_I is the period after rationally blowing down the X_i for $i \in I$: the lens faces $a = \frac{1}{4}$ are Wahl degenerations, and at a corner the piece is $\mathbb{P}(1, 1, 4) \rightarrow \mathbb{CP}^2$ [V].
 - Stretching along $S^2 \times S^1$ gives $v \cdot U = 0$, by the Kronheimer–Mrowka/Friedman–Morgan argument.
 - $\text{Scal} \geq 0$ is genuinely needed, because the argument requires empty moduli spaces, not vanishing counts.
- (3) *Negative definiteness*. On negative pieces $v^2 \leq -\frac{1}{2}(h^2 + h'^2)$; with a general V in place of ϕ this follows by Cauchy–Schwarz, and $\Lambda_V^2 = \sigma_V$ is attainable because the closed piece is diagonal by Donaldson’s theorem [V].

Lemma 5 (family adjunction inequality; notes/T1 Lemma 3.1; [V] as a lattice statement, and sharp). *Let Γ be a segment of consecutive pieces carrying an abelian configuration, including Uhlenbeck limits, separated lens caps and halves at cut necks. Let $s^-(\Gamma)$ be the number of insertions on Γ not adjacent to positive pieces, and $m(\Gamma)$ the number of positive pieces. Then*

$$\kappa_\Gamma \geq \frac{1}{2}(s^-(\Gamma) + m(\Gamma)).$$

Globally, a class v with $T(v)$ insertions met only through bubbles satisfies $-v^2/4 + T(v) \geq \frac{n-m}{2} - C$.

Since $\kappa = \frac{n+3m}{8} + C'$, no Seiberg–Witten class has nonnegative level once $3n - 7m > C_0$. Broken limits can isolate any segment, so a *local* density condition is needed; one endpoint count gives the following.

- Broken fixed limits: excess $\geq (3k - 7)m - 3k + 5$, so spacing $k \geq 3$ suffices.
- Mixed limits, priced by the $\text{SO}(3)$ -monopole index: excess $\geq (k - 4)m - k + 2$, so spacing $k \geq 4$ is needed, and the bound is attained at $k = 3$ [V].

This one lemma replaces the manuscript’s fixed-run, mixed-run and component-score lemmas.

8. Proof of Theorem 1

Suppose V, V' exist (for the cosmetic case, $V = \phi$ and $V' = \phi^{-1}$). Form X_M with positive pieces spaced four apart and pads of length p , all chosen before M . Then

$$n_D = \frac{M - p}{8} + c_1, \quad 3n - 7m = 5M + 3p.$$

For M large and divisible by the exponent of $\text{GL}_r(\mathbb{Z}/2^N)$, we get $n_D \geq 1$ and $3n - 7m > C_0$. By Theorem 4 and Lemma 5, $2^{n_D-1}\Omega(X_M) = 0$, whereas $\Omega(X_M) \equiv \pm q \not\equiv 0 \pmod{2^N}$. This is a contradiction. The coefficients must be $\mathbb{Z}$ or $\mathbb{Z}_{(2)}$: over $\mathbb{F}_2$ the relation is empty once $n_D \geq 2$, which is why B has to be integral.

Where the cosmetic map enters. We need $\frac{1}{5} < m/n$ (so that $n_D \rightarrow \infty$) and $m/n < \frac{3}{7}$ (so that the Seiberg–Witten side is empty), together with spacing 4. Without V , every B must be followed by H , because the reversed interval $Y_{-2} \rightarrow Y_2$ has $b^+ = 2$. That forces spacing 1, and then Seiberg–Witten classes do contribute. This is consistent with $HB \simeq 1 \pmod{2}$.

9. Assessment

- *Is it alien?* No. The ingredients are DLME and CDX triangles, Donaldson's μ -classes, Kronheimer–Mrowka nonvanishing, Feehan–Leness/Pidstrigach–Tyurin localization, and $b^+ = 1$ wall-crossing on rational surfaces. The genuinely new idea is to use the $\mathrm{SO}(3)$ -monopole cobordism as a *vanishing theorem for a secondary invariant*, with the cosmetic map arranging the ratio of (-4) -spheres to b^+ .
- *Why the manuscript reads as alien.*
 - Theorem 4 and Lemma 5 are never stated as such.
 - Classical objects are replaced by bespoke ones: holonomy sweeps for μ -classes, explicit toric estimates for wall-crossing, case-by-case dimension counts for one inequality.
 - The argument is run on the capped 4-manifold rather than in Floer homology.
- *A Floer-theoretic (DLME-shaped) proof.* One would want an S^1 -equivariant $\mathrm{SO}(3)$ -monopole Floer theory of $(Y_{\pm 2}, t)$, spin-graded, with localization weights 2^{n_D-1} , in which w lowers ℓ_t by $\frac{1}{8}$ per period.
 - Monotonicity cannot hold for single cobordisms: $HB \simeq 1 \bmod 2$ has spin shift $-\frac{1}{2}$.
 - For blocks with irreducible Floer ends it fails because of end terms at the 3-dimensional Seiberg–Witten points. Y_2 is not an L-space, since $2 < 2g - 1$. Only a version including the central flats is viable, at 20–35%.
 - No such theory exists. The capped construction is precisely the device that avoids it ([notes/B2](#), [notes/B3](#)).
- *Consistency checks.*
 - At ± 1 the same scheme reduces to DLME's energy argument, since $\kappa < 0$ [V].
 - For $\pm p$ with $p \geq 3$ it fails at the outset, because of non-central $U(1)$ flats [V].
 - With spacing 1 (no cosmetic map) the exclusions provably fail, as they must.
- *Risk.* Overall confidence that the reorganized argument is a correct proof is 35–45%. The weak points are:
 - the projection for mixed limits;
 - uniformity of the chamber condition at the faces of the cube;
 - persistence of the $\mu(S)$ representatives under Uhlenbeck limits, which Donaldson's divisor V_S should settle;
 - the lens cancellation in the coupled problem.